

Lecture 1: An Overview of Credit Market and Credit Analytics

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What is credit market?

Definition: Credit market refers to the market through which companies and governments issue debt to investors, such as commercial papers, government bonds, corporate bonds, securitized products. It is also called *debt capital market* or simply *debt market* or *bond market*.

Another market is called *equity market* or *stock market*

- ▶ They are related to each other. In the extreme case, when default occurs, the equity value is zero while bonds have a residual value or a recovery value.
- ▶ The credit market dwarfs the equity market in terms of dollar value and complexities.
- ▶ Credit market sometime is referred to as the **canary in the mine**, credit market typically shows signs of distress before the equity market.
- ▶ Before the financial crisis, a large portion of Wall Street professionals were working in credit market.
- ▶ Insurance company has most of its investments in debt market securities.

What are credit market instruments?

- ▶ commercial papers (CP)
- ▶ government bonds such as US government bonds: treasury bills, treasury notes, and treasury bonds
- ▶ corporate bonds including convertible bonds, *coco* etc. Investment grade bonds and speculative grade bonds or *junk bonds*.
- ▶ commercial mortgage loans (CML), residential mortgage loans (RML)
- ▶ securitized products: (cash) collateralized debt obligations (CDO), asset backed securities (ABS) (subprime mortgages, student loans, small enterprise loans, movie rights, aircraft leasing, infrastructure projects, cellular towers etc), collateralized loan obligations (CLO), commercial mortgage-backed securities (CMBS), RMBS
- ▶ credit derivatives: single name credit default swap (CDS), small basket, portfolio credit default swap (CDX), CDX tranche, (synthetic) CDO or single tranche CDOs, CDS on ABS bonds (PAUG CDS), CLO tranche

Credit Analytics

Definition: Analytic methods and tools used to study credit market

- ▶ Duration, convexity and immunization, less precise and very much introduced in a period of **flat yield curve** environment.
- ▶ Academic textbook and teaching ignore practical details, such as day count and day count rolling convention, different benchmark yield curves, these cannot be used for trading, hedging and risk management, such as risk aggregation
- ▶ Since the beginning of 1990s, Wall Street has taken credit not only as a type of risk, but also as a fundamental factor to trade on. This has introduced a set of new analytic thinking and methods which have not been covered by most of the textbooks.
- ▶ Credit derivative valuation has become the most major innovation before the financial crisis. Too complicated credit derivative products disappeared but most of the essential credit derivative products are still in use on a daily basis.

Credit Risk Management

Definition: Credit risk refers to the probability of loss due to a borrower's failure to make payments on any type of debt. Credit risk management is the practice of assessing individual credit defaults, and portfolio loss and its impact on the adequacy of a bank's capital and loan loss reserves.

1. Rating and score card have been used to assess individual borrower's risk
 - ▶ Rating and Score cards
 - ▶ Altman Z-score
 - ▶ KMV EDF
2. Modern credit portfolio management started from early 1990's. JP Morgan published the **CreditMetrics Technical Document** in 1997.
 - ▶ CreditManager, CreditPortfolio View, CreditRisk+, Tom Wilson's approach
3. Rating vs Pricing: Rating is a discrete pricing, while pricing is continuous rating. Pricing and valuation needs to consider much more details.

Corporate Bonds

- ▶ They enable firms to borrow money directly from the public
- ▶ Issued by large corporations as a cheaper alternative to bank financing

Structure

- ▶ similar to Treasury Bonds: pay semi-annual coupon
- ▶ rated by Moody's, Standard & Poor's, Duff and Phelps, Fitch Investor Services
- ▶ initial issue: by investment banks, through the underwriting process
- ▶ traded through bond dealers

Who buys them? insurance companies, mutual funds, hedge funds

Corporate Bonds

Riskiness

- ▶ default risk makes them much riskier than T-Bonds
- ▶ junk bonds: below-investment grade corporate bonds particularly risky

Bond Features

- ▶ Important features that bond issuers can set: covenants, options, cash flow patterns (via principal and coupon schedules), maturity, price, and rating

Corporate Bonds

- ▶ Bond covenants: rules that specify the rights of the lender and the restrictions on the borrower
 1. Asset covenants govern firms' acquisition, use, and disposition of assets. Specifies the rights bond holders have in the event of default (senior, junior, secured debt).
 2. dividend covenants: restrict the payment of dividends. (Why is this good for bond holders?)
 3. financing covenants: prevents firm from issuing new debt, diluting current bond holders' claims.
 4. bonding covenants: mechanism for enforcement of the covenants, usually an appointed trustee to oversee that the covenants are not violated
 5. financial ratio covenants
 6. sinking fund covenants: certain portion of the bonds must be retired by certain dates.

Corporate Bonds

Corporate bonds are rated according to their default risk. Credit ratings:

- ▶ Moody's: Aaa, Aa, A, Baa, Ba, B, Caa, Ca, C, D[1, 2, 3]
- ▶ S&P: AAA, AA, A, BBB, BB, B, CCC, CC, C, D[+, −]
- ▶ Investment grade ($\geq BBB$ or Baa) and “Junk Bonds” ($< BBB$ or Baa)
- ▶ Junk bond market has thrived over the last three decades growing from \$25 billion in 1980 to over \$147 billion new issues in 2009.

Classification of Corporate bonds:

- ▶ Secured bonds: specific collateral backing in the event of default.
- ▶ Debentures: unsecured bonds.
- ▶ Subordinated debentures: lower priority claim in the case of default than debentures.

Options attached to bonds:

- ▶ Callability: allows the issuing firm to retire the bonds before maturity at a prespecified price
 1. protects firms from bondholders who refuse to renegotiate covenants
 2. allows firm to retire bonds when interest rates fall substantially, or creditworthiness improves
 3. enables firms to institute sinking fund provisions
 4. consequently, callable bonds sell at a discount to non- callable bonds
 5. nearly all long-term corporate bonds used to be callable
 6. now, only the least creditworthy are, because derivatives and financial engineering have provided substitutes such as the interest rate swap option

Options attached to bonds:

- ▶ Convertibility: allows bondholder the option to convert the bond into another security, typically common stock.
- ▶ Exchangeability: allows issuing firm the right to exchange the bond for a different type of bond.
- ▶ Putability: allows bondholder to sell the bond back to the firm, consequently putable bonds sell at a premium to non-putable bonds (example: poison puts).

Corporate Bonds

Cashflow patterns:

- ▶ Straight coupon bond (bullet bond): fixed rate, semiannual coupon, principal paid only at terminal date (balloon payment)
- ▶ Zero coupon or pure discount bond: pay no periodic interest and have a single payment at maturity.
- ▶ Deferred coupon bond: interest is avoided for a certain period, so that cash-constrained firms can fund other projects.
- ▶ Perpetuity bonds (consols): bonds that last forever and therefore only pay interest.
- ▶ Annuity bond: bonds that pay a mix of interest and principal for a finite amount of time.

Bond Analytics Fixed income analysis is the process of determining the value of a debt security based on an assessment of its risk profile, which can include interest rate risk, default risk -risk of the issuer failing to repay the debt, market supply and demand for the security, call provisions and macroeconomic considerations affecting its value.

Major components:

- ▶ interest rate
- ▶ credit spread
- ▶ duration and convexity
- ▶ supply and demand
- ▶ macroeconomics/inflation

Bond Analytics (Classical Approach)

Suppose we have a bond which pays the coupon amount C every six months, and the final face value at the maturity in n years, the current market price is P

$$P = \frac{C}{(1 + y/2)} + \frac{C}{(1 + y/2)^2} + \dots + \frac{C + F}{(1 + y/2)^{2n}}.$$

For a given P , we can solve for y which is called *yield to maturity*. YTM gives you an average of yearly return if you hold the bond to the maturity.

Duration is defined as the weighted average of cash flow payment times(measured in number of years), where the weights are the present values of cash flows over the total value, which is the bond price.

$$D = \sum_{i=1}^n w_i t_i, w_i = \frac{C_i / (1 + y/2)^{2t_i}}{P}$$

Bond Analytics (Continue)

It can be shown that

$$\frac{dP}{dy} = -\frac{PD}{1 + y/2}$$

If we define

$$D^* = \frac{D}{1 + y/2}$$

as *modified duration*. Then we would have

$$D^* = -\frac{1}{P} \frac{dP}{dy}$$

This shows that the modified duration is a measure for bond price sensitivity with respect to the change in interest rate.

Bond Analytics (Continue)

Convexity

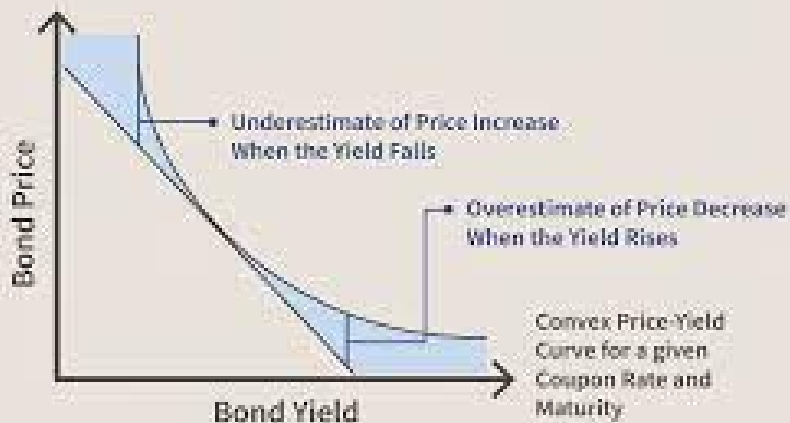
$$C = \frac{1}{P} \frac{d^2 P}{dy^2}$$

$$\frac{\Delta P}{P} \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2$$

Uses of duration and convexity

- ▶ Duration is used as an approximate answer to the impact of interest rate changes on the bond price.
- ▶ Due to the positive convexity, using duration only tends to: underestimate of price increase when yield falls, and overestimate of price decrease when yield rises.
- ▶ trading strategies: long duration: make money when interest rates are falling, which usually happens during recession; short duration: make money when interest rates are increasing.

Bond Analytics (Continue)



Interest Rate Immunization

Interest rate immunization: a portfolio management strategy designed to take advantage of the offsetting effects of interest rate risk and reinvestment risk or asset and liability so that net position is "immunized" from the change of interest rates.

$$\begin{aligned} S &= A - L \\ \Delta S &= \Delta A - \Delta L \\ &\cong (LD_L - AD_A)\Delta Y + \frac{1}{2}(AC_A - LC_L)(\Delta Y)^2 \end{aligned}$$

- ▶ If initially asset value and liability value are equal, their durations are also equal, and the convexity of asset is larger than the convexity of liability, the surplus would be positive if interest rate moves
- ▶ This is based on the *parallel shift* of the yield curve, and the yield curve is assumed to be flat.

Bond Analytics (Modern Approach)

Issues with the traditional bond analytics:

- ▶ YTM is a measure associated with each bond, it depends on the coupon, maturity, bond price. It is hard to aggregate, for example, bond portfolio YTM or duration;
- ▶ This traditional calculation does take consideration of the actual market convention. For example, bond coupon is paid twice of a year, but the actual coupon paid is not exactly $1/2$ of the annual coupon. it depends on the **day count conventions**.
- ▶ There are many different curves associated with the default risk of the issuers. In US we have Treasury curve, LIBOR curve, Muni Curve. In China we have government bond curve, we have bond curve from CDB.
- ▶ There is no such thing called a constant interest rate, most of the times we will have different YTM for bonds of different maturities, this gives the **yield curve** or **term structure of interest rates**.

$$P = c \times F \times \Delta_1 D(0, t_1) + c \times F \times \Delta_2 D(0, t_2) + \dots + (c \times F \times \Delta_m + F) D(0, t_m)$$

where

- ▶ c is annual coupon rate, F is face value
- ▶ $t_1, \dots, t_m$ are cash flow payment dates
- ▶ $\Delta_1, \dots, \Delta_m$ coupon payment day count fractions which could be 30/360 or 30/365.
- ▶ $D(0, t_i)$ is the current discount factor (to the valuation date or bond traded date) for one dollar payment at time t_i .

You need to first calculate the coupon payment dates (day count rolling convention), and then the coupon amount for each coupon payment period,

Term Structure of Interest Rates- Yield curve

The yield curve can be upward sloping (typical), hump-shaped, or flat once we graph the bond YTM as a function of time to maturity. We first study how to express an yield curve. (1) We first define a zero rate from which we can calculate the accumulation factor and discount factor

$$C(0, t) = e^{r(t)t}; Z(0, t) = e^{-r(t)t}$$

$$r(t) = -\frac{1}{t} \ln Z(0, t)$$

(2) Forward discount factor

$$Z(0, t_2) = Z(0, t_1)Z(0, t_1, t_2), Z(0, t_1, t_2) = \frac{Z(0, t_2)}{Z(0, t_1)}$$

Forward Rate

$$Z(0, t_1, t_2) = e^{-f(0, t_1, t_2)(t_2 - t_1)}$$

If you know the one year and two year zero rates, how do you get the one year interest rate in one year? Suppose you have one dollar, and you can have two investment strategies: A. Invest for 2 years and receive $e^{r(t_2)t_2}$ B. Invest for one year first, and then for another one year $e^{r(t_1)t_1} e^{f(0, t_1, t_2)(t_2 - t_1)}$

$$f(0, t_1, t_2) = \frac{r_2 t_2 - r_1 t_1}{t_2 - t_1} = \frac{\ln(Z(0, t_2)) - \ln(Z(0, t_1))}{t_2 - t_1}$$

Instantaneous forward rate:

$$f(t) = -\frac{d}{dt} \ln Z(t) = \frac{d}{dt} r(t)t$$

Forward Rate

We can also obtain discount factor from instantaneous forward rate

$$Z(t) = \exp \left(- \int_0^t f(s) ds \right), \frac{r_2 t_2 - r_1 t_1}{t_2 - t_1} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} f(s) ds$$

This shows that the average of forward rate over any time interval is equal to the discrete forward rate for that interval.

$$r_t t = r_{t_{i-1}} t_{i-1} + \int_{t_{i-1}}^t f(s) ds$$

This shows the relationship between spot rate and forward rate. We can express an yield curve in terms of discount factors, accumulation factors, zero rates (spot curve) or forward rates.

Curve Fitting and Bootstrapping

There exist quite a few **yield curve fitting methods**. The purpose is to find a simple representation of the whole yield curve, such as B-Spline, Nelson & Siegal approaches. These methods are mainly used for economic research. There are also some more complicated approach to yield curve fitting such as two plus model used by Salmon for bond arbitrage, in which two stochastic Brownian motions are assumed to be the driving factor. These approaches usually do not require EXACT fitting to match every traded bond price. Another approach is called **yield curve bootstrapping** in which a series of traded bond prices are used, and it tries to reprice every used instrument. In addition it might require some kind of smoothness of the yield curve. It is better to apply some simple assumption on the forward curves instead of the discount factors or zero rates. Constant flat forward over a small time interval has been used a lot in practice.

Theory of Term Structure

- ▶ (Pure) Expectations Hypothesis: forward rates equal expected future interest rates; a rising yield curve is due entirely to expectations of higher future interest rates; the yield curve is often rising when: the Fed is lowering interest rates to stimulate the economy; people foresee higher future interest rates due to growth; this hypothesis is not generally true in practice.
- ▶ Liquidity Premium Hypothesis: the yield curve is rising because most investors have short horizons; this hypothesis is not generally true in practice.
- ▶ Fama and Bliss: Fama and Bliss found empirical evidence that the higher the term premium of a default-free bond, the higher its expected short-term return; some mutual funds and hedge funds trade on this finding

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